

Section 3 Probability

Statistics I · Fall 2026 · Zong-Wei Yeh · Department of Finance, CCU

Overview

Sections 1 and 2 described data we already hold. But we sample in order to say something about a population we cannot see. Every such statement can be wrong. To measure that chance we need a language for uncertainty. That language is probability. This section builds it step by step, starting from three axioms.

This section has four parts.

- §3.1 names the objects probability talks about: experiments, outcomes, events. It then states the three axioms that every probability must obey. The working rules follow from them by proof.
- §3.2 computes probabilities in the simplest setting, where all outcomes are equally likely. That setting turns probability into counting.
- §3.3 asks how the probability of one event changes once another is known. This is conditional probability. It makes probability useful for inference.
- §3.4 reverses that reasoning, from effect to cause, with Bayes' theorem.

Sections 4 to 7 are built on these four ideas. Random variables, distributions and the sampling distributions of Section 7 all rest on them.

3.1 Random Experiments, Events, and the Axioms of Probability

Definition 3.1 (Random experiment, sample space, event).

- (i) A random experiment (隨機實驗) is a process with two or more possible outcomes. Which outcome occurs is _____.
- (ii) The sample space (樣本空間) S is the set of all basic outcomes (基本結果). The basic outcomes are defined so that _____ of them occurs.
- (iii) An event (事件) is any subset $A \subseteq S$. We say A occurs if the outcome that happens _____ A .

A random experiment is the situation itself, such as rolling a die or observing tomorrow's market move. The sample space lists everything that could happen. An event is a question we can ask about the outcome, such as "was it even?". We write each event as a set of basic outcomes.

Table 1 gives five examples. Each row names an experiment, lists its sample space, and writes one event as a set. In the first two rows, H means head and T means tail. In the last row, U means that the stock price rises on that day. N means that it does not rise. So UN means a rise on day 1 and no rise on day 2. Suppose the outcome is UN . Then the event "rises on day 1" occurs. The event "never rises", which is $\{NN\}$, does not occur.

Random experiment	Sample space S	One event, as a set
Toss a coin once	$\{H, T\}$	head: $\{H\}$
Toss a coin twice	$\{HH, HT, TH, TT\}$	at least one head: $\{HH, HT, TH\}$
Roll a die	$\{1, 2, 3, 4, 5, 6\}$	even: $\{2, 4, 6\}$
A firm bids on 5 contracts and counts its wins	$\{0, 1, 2, 3, 4, 5\}$	at least 4 wins: $\{4, 5\}$
A stock over two trading days	$\{UU, UN, NU, NN\}$	rises on day 1: $\{UU, UN\}$

Table 1: Five random experiments and their sample spaces. The last column gives one event for each experiment, first in words and then as a set of basic outcomes. Two tosses of a coin give four basic outcomes, not two, because each outcome records both tosses.

One experiment can have more than one sample space. The choice depends on what we record. For example, roll two dice. If we record the pair of faces, such as $(2, 5)$, there are 36 basic outcomes. If we record only the sum, the sample space is $\{2, 3, \dots, 12\}$. Both sample spaces are correct. If the dice are fair, the outcomes are equally likely only in the first one. Example 3.11 in §3.2 shows why this matters.

Events are *sets*, so we combine them with set operations.

Definition 3.2 (Set operations; mutually exclusive; exhaustive).

For events $A, B \subseteq S$:

- (i) the union (聯集) $A \cup B$ occurs if A or B (or both) occur;
- (ii) the intersection (交集) $A \cap B$ occurs if A and B both occur;
- (iii) the complement (餘事件) A^c occurs if A does not occur;
- (iv) A and B are mutually exclusive (互斥) if $A \cap B = \emptyset$, the empty event (空事件). They cannot both occur. More generally, a list of events $E_1, E_2, \dots$ (finite or not) is mutually exclusive if every pair in it is;
- (v) events $E_1, \dots, E_k$ are collectively exhaustive (周延) if $E_1 \cup \dots \cup E_k = S$, that is, one of them must occur.

A list of events that is both mutually exclusive and collectively exhaustive is called a *partition* (分割) of S . We will need it in §3.4. Figure 1 shows the three basic operations.

Here is the stock from Table 1 again, with $S = \{UU, UN, NU, NN\}$. Let A be “rises on day 1” and let B be “rises on day 2”. So $A = \{UU, UN\}$ and $B = \{UU, NU\}$. The definitions give the following events.

- $A \cup B = \{UU, UN, NU\}$. The stock rises on at least one of the two days.
- $A \cap B = \{UU\}$. The stock rises on both days.
- $A^c = \{NU, NN\}$. The stock does not rise on day 1.
- Let $C = \{NN\}$, the event “never rises”. Then $A \cap C = \emptyset$, so A and C are mutually exclusive. The stock cannot rise on day 1 and also never rise.
- The events $\{UU\}$, $\{UN, NU\}$ and $\{NN\}$ mean “rises on two days”, “rises on one day” and

“rises on no day”. They are mutually exclusive. Their union is S . So they form a partition of S .

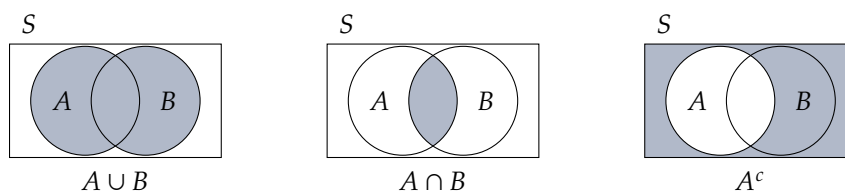


Figure 1: The three set operations on events, shaded. $A \cup B$ is everything in either circle. $A \cap B$ is the overlap. A^c is everything in the sample space S outside A . Such a picture is the quickest way to check a rule. For example, the addition rule below must not count the overlap twice.

To turn “how likely” into arithmetic, we require every probability to obey three rules. Everything else is a theorem.

Definition 3.3 (Axioms of probability).

A probability assigns to each event $A \subseteq S$ a number $P(A)$ such that the three axioms (機率公理) below hold.

(A1) $P(A) \geq 0$;

(A2) $P(S) = \underline{\hspace{1cm}}$;

(A3) if $A_1, A_2, \dots$ are mutually exclusive, then $P(\bigcup_i A_i) = \underline{\hspace{1cm}}$.

We use axiom (A3) more than the other two. It is the rule behind almost every calculation in this section. The first two axioms only fix the scale. Example 3.4 applies the axioms to the stock of Table 1.

Example 3.4. Return to the stock over two trading days, with $S = \{UU, UN, NU, NN\}$. Suppose $P(\{UU\}) = 0.3$, $P(\{UN\}) = 0.2$, $P(\{NU\}) = 0.2$ and $P(\{NN\}) = 0.3$. (a) Use the axioms to find the probability that the stock rises on day 1. Then find the probability that it rises on at least one day. (b) Explain why the four numbers 0.4, 0.3, 0.2, 0.2 cannot be the probabilities of $\{UU\}$, $\{UN\}$, $\{NU\}$ and $\{NN\}$.

Everything else follows from the three axioms. The next two propositions are the first consequences. We use them constantly.

Proposition 3.5 (Complement rule).

For any event A , $P(A^c) = 1 - P(A)$. In particular $P(\emptyset) = 0$ and $0 \leq P(A) \leq 1$.

Proof.

□

Proposition 3.6 (Addition rule).

For any events A, B , $P(A \cup B) =$.

Proof.

□

Example 3.7. Roll one fair die, so $S = \{1, 2, 3, 4, 5, 6\}$ with equally likely outcomes. Let $A = \{\text{even}\} = \{2, 4, 6\}$ and $B = \{1, 2, 3\}$. Find $A \cup B$, $A \cap B$, A^c , and their probabilities.

Example 3.8. Among a firm's investors, $P(\text{holds stocks}) = 0.5$, $P(\text{holds bonds}) = 0.4$, and $P(\text{holds both}) = 0.3$. Find the probability an investor holds stocks or bonds, and the probability they hold neither.

The addition rule also gives a bound for the union of many events.

Proposition 3.9 (Boole's inequality).

For any events $A_1, \dots, A_n$, mutually exclusive or not,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \quad .$$

Proof.

□

Boole's inequality (布爾不等式) gives an upper bound for the probability that at least one of several events occurs. It needs no information about how the events overlap. That makes it useful, because the overlaps are often unknown. When the events are mutually exclusive, the bound is exact: that case is axiom (A3). The proof above uses only the axioms and the addition rule. So Boole's inequality is a theorem, not a fourth axiom.

3.2 Classical Probability and Counting

In the oldest setting, every outcome is equally likely. Dice, cards and random sampling all work this way.

Definition 3.10 (Classical probability).

Suppose the sample space S has N equally likely outcomes, and event A contains N_A of them. Then the classical probability (古典機率) of A is

$$P(A) = \frac{N_A}{N}.$$

Example 3.11. Roll two fair dice, one red and one blue. (a) Find the probability that the two faces add up to 7. (b) A student uses the sample space $S = \{2, 3, \dots, 12\}$ of possible sums. It has 11 outcomes. The student applies the definition of classical probability to this sample space. What answer does the student get? What is wrong?

With equally likely outcomes, computing $P(A)$ becomes a pure counting problem. In Example 3.11 we could still list all 36 outcomes. Usually the list is far too long. So we need ways to count without listing. Three tools are enough.

The multiplication principle (乘法原理). Suppose a task is done in stages. There are n_1 choices at the first stage, n_2 at the second, and so on up to n_k . The number of choices at each stage does not depend on what was chosen earlier. Then the number of ways to complete the whole task is

The other two counts are built on this one.

Definition 3.12 (Permutations and combinations).

Here $n!$ (read “ n factorial”) means $n(n-1)\cdots 2\cdot 1$, with $0! = 1$ by convention. From n distinct objects:

- (i) an ordered selection of x objects is called a permutation (排列). The number of permutations is

$$P(n, x) = \quad ;$$

- (ii) an unordered selection of x objects is called a combination (組合). The number of combinations is

$$C(n, x) = \binom{n}{x} = \quad .$$

Every time, we must decide whether _____. A ranking, such as gold, silver and bronze, is a permutation. A committee, where only membership counts, is a combination. The two counts are linked by $C(n, x) = P(n, x) / \underline{\hspace{1cm}}$. Each unordered set of x objects can be arranged in that many ways. The combination $C(n, x)$ returns in Section 5, where it counts the success patterns of the binomial distribution.

Example 3.13. A password is two letters (A–Z) followed by three digits (0–9), with repeats allowed. How many passwords are possible?

Example 3.14. From 5 analysts, (a) how many ordered rankings of the top 3 are there? (b) how many 3-person committees?

Example 3.15. A committee of 3 is chosen at random from 6 men and 4 women. Find the probability it contains exactly 2 women.

3.3 Conditional Probability, Independence, and the Multiplication Rule

Inference is about revising what we believe as information arrives. The tool for that is conditional probability. It is the probability of A once we know that B has occurred.

Definition 3.16 (Conditional probability).

For events A, B with $P(B) > 0$, the conditional probability (條件機率) of A given B is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Knowing that B has occurred shrinks the _____ from S to B . Example 3.17 shows this on a die.

Example 3.17. Roll one fair die. Let $A = \{4, 5, 6\}$, and let B be the event that the outcome is even. Find $P(A)$ and $P(A | B)$. Explain why they differ.

Rearranging the definition gives a rule for the probability that *both* events happen.

Proposition 3.18 (Multiplication rule).

For events with positive probability,

$$P(A \cap B) = \quad .$$

Proof.

□

The multiplication rule (乘法法則) builds the probability of a sequence one stage at a time. That makes it ideal for draws _____. One special case matters a great deal: when knowing B tells us nothing about A .

Definition 3.19 (Independence).

Events A and B are independent (獨立) if

$$P(A \cap B) = \quad ,$$

and equivalently, when $P(B) > 0$, if $P(A | B) = P(A)$.

So when the two events are independent, conditioning changes nothing. Independence is

something to be checked, not taken for granted. Section 7 assumes independent samples. So this check matters there.

Example 3.20. Of 100 investors, cross-tabulated by whether they hold stocks (A) and bonds (B):

	B	B^c	total
A	30	20	50
A^c	10	40	50
total	40	60	100

Find $P(B | A)$ and $P(A | B)$. Are A and B independent?

Example 3.21. Two cards are drawn without replacement from a standard 52-card deck. Find the probability that both are aces.

Example 3.22. A circuit uses two bulbs. Each bulb fails with probability 0.1. The two bulbs behave independently of each other. (a) The bulbs are wired in *series*. The circuit lights only if both bulbs work. Find the probability that the circuit lights. (b) The bulbs are wired in *parallel*. The circuit lights if at least one bulb works. Find the probability that the circuit lights. (c) Which wiring is more reliable, and why?

3.4 Total Probability and Bayes' Theorem

Often we know the probability of an effect under each of several causes. Then we want one of two things. Sometimes we want the overall probability of the effect. More often we want the probability of a cause given the effect, which is harder and more useful.

Theorem 3.23 (Total probability).

Let $B_1, \dots, B_k$ be mutually exclusive and collectively exhaustive events, each with $P(B_i) > 0$. Then for any event A ,

$$P(A) = \sum_{i=1}^k P(A|B_i)P(B_i).$$

Proof.

□

A _____ divides the sample space into cases. Each outcome falls in exactly one case. So no case is counted twice, and none is left out. Total probability (全機率) answers the forward question. It gives the overall probability of A from what we know inside each case. Reversing the conditioning gives Bayes' theorem.

Theorem 3.24 (Bayes' theorem).

Let $B_1, \dots, B_k$ be mutually exclusive and collectively exhaustive events, each with $P(B_i) > 0$. Let A be an event with $P(A) > 0$. Then

$$P(B_j | A) = \frac{P(B_j)P(A|B_j)}{P(A)}.$$

Proof.

□

Bayes' theorem (貝氏定理) inverts the conditioning. We can usually measure $P(\text{effect} | \text{cause})$. But we usually want $P(\text{cause} | \text{effect})$. Bayes' theorem turns the first into the second. The prior probability (事前機率) $P(B_j)$ is called the _____. Most people forget it. Forgetting it produces the surprising answer in Example 3.26.

Example 3.25. Factory 1 makes 60% of a firm's chips, with a 2% defect rate. Factory 2 makes the other 40%, with a 5% defect rate. (a) What fraction of all chips are defective? (b) Given a defective chip, what is the probability it came from factory 2?

Example 3.26. A disease affects 1% of a population. A test is 99% sensitive (positive if diseased) with a 5% false-positive rate (positive though healthy). A randomly chosen person tests positive. What is the probability they actually have the disease?

Example 3.27. A survey asks whether the respondent has ever cheated on an exam. To protect privacy, each respondent tosses a fair coin in private. Nobody but the respondent sees either coin. On heads, the respondent tosses a second fair coin. The answer is “yes” if the second coin shows heads, and “no” if it shows tails. On tails, the respondent answers the cheating question truthfully. The tosses do not depend on whether the respondent has cheated. Let π be the proportion who have cheated. (a) Find $P(\text{yes})$ in terms of π . (b) In a large survey 40% answer “yes”. Estimate π . (c) One respondent answers “yes”. Find the probability that this person has cheated.

Exercises

Exercise 3.1. A fair die is rolled. Write the sample space S , the event $A = \text{"the outcome is odd"}$, and find $P(A)$ and $P(A^c)$.

Exercise 3.2. Events A, B satisfy $P(A) = 0.6$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$, $P(A^c)$, and $P(A^c \cap B^c)$.

Exercise 3.3. From 8 funds, (a) how many ordered "top two" rankings are there? (b) how many 3-fund portfolios (order irrelevant)?

Exercise 3.4. An urn holds 5 red and 3 blue balls. Two are drawn at random without replacement. Find the probability that both are red.

Exercise 3.5. Of 200 customers, 90 bought online and 110 in store. Of the online buyers 60 paid by card; of the in-store buyers 40 paid by card. Find $P(\text{card} \mid \text{online})$ and $P(\text{online} \mid \text{card})$.

Exercise 3.6. Events A, B satisfy $P(A) = 0.5$, $P(B) = 0.6$, $P(A \cap B) = 0.3$. Are they independent? Justify.

Exercise 3.7. A box has 10 items, 3 of them defective. Two are drawn without replacement. Find $P(\text{both defective})$ and $P(\text{neither defective})$.

Exercise 3.8. Of all incoming email, 40% is spam. The word "free" appears in 90% of spam messages and in 10% of legitimate messages. An incoming email contains "free". Find the probability it is spam.

Exercise 3.9. Events A and B satisfy $P(A) = 0.7$ and $P(B) = 0.5$. Nothing else is known about them. Show that $0.2 \leq P(A \cap B) \leq 0.5$.

Exercise 3.10. (a) Apply the addition rule twice to show that, for any events A, B, C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

(b) Among a bank's clients, 50% hold stocks, 40% hold bonds and 30% hold funds. Also, 20% hold stocks and bonds, and 15% hold stocks and funds. In addition, 10% hold bonds and funds. Finally, 5% hold all three. (Each of the pairs includes the clients who hold all three.) A client is chosen at random. Find the probability that the client holds at least one of the three. Then find the probability that the client holds none of them. Last, find the probability that the client holds exactly one of them.

Exercise 3.11. Five people are chosen. Each was born on one of the 7 days of the week. Assume that all 7^5 possible lists of five weekdays (one per person, in order) are equally likely. Find the probability that at least two of the five were born on the same day of the week.

Exercise 3.12. (a) Suppose A and B are independent. Prove that A and B^c are independent. Then prove that A^c and B^c are independent. (b) Suppose A and B are mutually exclusive, with $P(A) > 0$ and $P(B) > 0$. Prove that A and B are not independent. (c) A firm uses two suppliers. Supplier 1 fails to deliver with probability 0.1, and supplier 2 with probability 0.2. The two failures are independent. Find the probability that neither supplier fails, and the probability that at least one fails.

Exercise 3.13. (a) Show that for events A, B, C with $P(A \cap B) > 0$,

$$P(A \cap B \cap C) = P(A) P(B | A) P(C | A \cap B).$$

(b) A box holds 12 items, and 4 of them are defective. Three items are drawn one at a time without replacement. Find the probability that all three are defective. Then find the probability that the first defective item appears on the third draw.

Exercise 3.14. A bond fund puts its bonds into three groups: investment grade (70% of the bonds), speculative (20%) and distressed (10%). Over one year, an investment-grade bond defaults with probability 1%. A speculative bond defaults with probability 10%, and a distressed bond with probability 40%. A bond is chosen at random. (a) Find the probability that it defaults within the year. (b) It defaults. Find the probability that it was distressed. (c) It does not default. Find the probability that it was investment grade.

Exercise 3.15. A game show has three closed doors. A car is behind one door, and a goat is behind each of the other two. Each door hides the car with probability $1/3$. You pick door 1. The host knows where the car is. The host then opens one of the other two doors. The host follows two rules. First, the door opened is never your door and never the car's door. Second, suppose both of the other doors hide goats. Then the host picks one of them at random, each with probability $1/2$. The host opens door 3 and shows a goat. Find the probability that the car is behind door 2. Should you switch to door 2?